

Problem. Suppose a bisector of angle C in a nondegenerate triangle ABC bisects the opposite side. Show that the triangle is isosceles.

Let D be the point of the intersection of the bisector with $[AB]$.

Assume, $AD = BD$ and
 $\angle ACD = \angle BCD$.

Note also that

$\triangle ACD$ and $\triangle BCD$ share side $[CD]$.

no, it is
ASS & $\nexists$ such
condition

By SAS we have

$$\triangle ACD \cong \triangle BCD.$$

In particular, $AC = BC$.

Therefore, $\triangle ABC$ is isosceles.